Alexander Marsh

groethe@umich.edu | LinkedIn | GitHub

EDUCATION

University of Michigan - Ann Arbor, *Bachelor of Science in Statistics, Data Science* | **In-Major GPA**: 3.848/4.000 August 2023 - May 2027

- **Awards / Honors:** University Honors (x3), EECS 183 Elevators Showcase 1st Place
- **Course Highlights:** Statistical Modeling and ML Techniques, Deep Learning, Natural Language Processing, Database Systems Management, Data Mining, Probability, Theoretical Statistics, Optimization Methods, Statistical Computing, Data Structures, Object Oriented Programming

King's College London, Informatics January 2026 - June 2026

EXPERIENCE

GEICO | Pricing Analyst Intern May 2026 - August 2026

- Led actuarial trend and loss development (LDF) analyses, querying complex loss data with **SQL** in **Snowflake** and modeling it in **R** and **Power BI** to produce data-driven rate proposals for senior stakeholders.
- Built **predictive models** for loss development tail factors, using Ward's hierarchical **clustering** to segment state coverage credibility complements.

University of Michigan Human Performance & Sport Science Center | Sports Analyst Assistant August 2026 - Present

- Collaborated directly with the UM Men's Basketball team to produce and maintain an **R Shiny** dashboard supporting player performance, providing in-depth player comparison, athleticism, and fatigue **time-series analysis**.
- Cleaned, processed, and engineered **Catapult** wearable sensor tracking data to answer coaching staff's analysis requests.

University of Michigan Department of Statistics | STATS 250 Instructional Assistant August 2025 - Present

- Support instruction for lectures of over **500** students, guiding groups through analysis, reviewing their work, and challenging their reasoning on inference techniques including **CI, hypothesis testing**, **ANOVA**, and **Chi-Square** tests.
- Break down statistical concepts into clear explanations for students of all backgrounds and grade group assignments for accuracy with timely, constructive feedback.

Michigan Sports Analytics Society (MSAS) | R Tutorial Lead & Lead Data Analyst August 2024 - Present

- Co-developed a **Python** and **R** analysis of March Madness Cinderella predictors, formally presented at the MSAS Data Analytics symposium.
- Lead **R** instruction on sports analytics data analysis, preparing members for professional and symposium presentations.

East Quad Blue Cafe | Manager August 2024 - Present

- Direct daily operations and coordinate team efforts, delegating tasks to maintain efficiency and service quality.
- Manage inventory and restocking, proactively resolving operational challenges to keep workflow smooth and up to standard.

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python, SQL, C++, R, Java

Tools: PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, SciPy, Snowflake, MySQL, PostgreSQL, MongoDB, Git/GitHub, Docker, AWS, Streamlit, R Shiny, Tableau, Power BI

Concepts: Machine Learning, Deep Learning, Large Language Models, Natural Language Processing, Feature Engineering, Predictive Modeling, Cross-Validation, Regularization, Clustering, Statistical Inference, Hypothesis Testing, A/B Testing

Certifications: Python Data Structures, Python Applied ML, Python Data Mining, Python Text Processing, Python Unsupervised Learning

PROJECTS

Insights for University of Michigan Men's Basketball | Link September 2025 - April 2026

- Partnered with *SkillCorner* on a large-scale analytics project, engineering spatial and spacing features from high-frequency **computer vision** player-tracking data in **Python** and **Tableau** to deliver actionable insights to the **Michigan Basketball** coaching staff, presented at the organization's annual Symposium.
- Built **XGBoost** models of Shot Quality and EPA to quantify possession-level value, interpreting results for coaching staff to assess pick-and-roll decision effectiveness.

Deep Neural Network for NBA Player Statistics Forecasting

- Designed a deep neural network in **PyTorch and TensorFlow**, trained on rolling seasons, to forecast NBA player performance.
- Prepared inputs using **data preprocessing techniques** such as normalization and one-hot encoding of rotational and positional features.
- Tuned hyperparameters and validated performance via **K-Fold cross-validation**, applying regularization in **Python** to reduce overfitting.

NBA Interactive Dashboard | Link May 2026

- Built an end-to-end interactive web application, version-controlled on **GitHub**, that ingests and processes live PBP data through the **nba_api**, generating matchup summaries from 2005–present.
- Modeled real-time game momentum shifts with time-decayed algorithms, smoothing data using **SciPy** (Pchip interpolation) and **Pandas**.
- Designed a reactive **Streamlit** front end, enabling seamless parameter manipulation and real-time visualization updates.

C++ Software Engineering Projects | Link

- Implemented greedy and optimized **C++** solutions to the Traveling Salesperson Problem, applying Furthest Neighbor, **2-opt**, and **Prim's algorithm** to compare route-construction strategies.
- Built a **C++** bank transaction system with credential validation and login handling, using **hash tables** for efficient queries and transaction storage.

Heliocentrism and Success | Link June 2025 - August 2025

- Defined and classified heliocentrism through EDA, then conducted **regression analysis** and **inference** in **R** and **Python** to evaluate its link to team success.
- Examined individual player case studies to show how specific load-based attributes contributed to team performance.

Wayne Country Legal Justice Analysis August 2025 - Present

- Built a Python pipeline (BeautifulSoup, pytest) that parses and standardizes 31,000 Wayne County HTML case files (2000–2025) into structured JSON and a SQLite database, using AI-assisted development
- Analyzed weapons felony charges against Michigan Compiled Laws (MCL) statutes to identify potential over-prosecution and charging disparities